7. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To develop a Java Servlet-based web application that stores user preferences (such as theme and language) using cookies and retrieves them on subsequent visits to personalize the weather information webpage.

ALGORITHM

1. **Create a Dynamic Web Project** in Eclipse or any IDE.
2. **Add Servlet API library** (Apache Tomcat server).
3. **Create an HTML form** to accept user preferences:
   Theme (Light/Dark)
   Language (English/Tamil/etc.)
4. **Develop a Servlet (PreferenceServlet)**:
   Read user inputs from the form.
   Store preferences using cookies.
5. **Develop another Servlet (DisplayServlet)**:
   Retrieve cookies.
   Display saved preferences along with weather info.
6. **Configure web.xml** (if not using annotations).
7. **Run the project on server** (Tomcat).
8. **Test in browser**:
   First visit → set preferences.
   Next visit → preferences auto-loaded.

SOURCE CODE

1. HTML Page (index.html)

```
<!DOCTYPE html>
<html>
<head>
    <title>Weather Preferences</title>
</head>
<body>
    <h2>Select Your Preferences</h2>
    <form action="PreferenceServlet" method="post">
        Theme:
        <select name="theme">
            <option value="Light">Light</option>
            <option value="Dark">Dark</option>
        </select>
        <br><br>

        Language:
        <select name="language">
```

```
            <option value="English">English</option>
            <option value="Tamil">Tamil</option>
        </select>
        <br><br>

        <input type="submit" value="Save Preferences">
    </form>

    <br>
    <a href="DisplayServlet">View Weather Page</a>
</body>
</html>
```

2. PreferenceServlet.java

```
import java.io.; import javax.servlet.; import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    String theme = request.getParameter("theme");
    String language = request.getParameter("language");

    // Create cookies
    Cookie themeCookie = new Cookie("theme", theme);
    Cookie languageCookie = new Cookie("language", language);

    // Set expiry time (1 day)
    themeCookie.setMaxAge(24 * 60 * 60);
    languageCookie.setMaxAge(24 * 60 * 60);

    // Add cookies to response
    response.addCookie(themeCookie);
    response.addCookie(languageCookie);

    PrintWriter out = response.getWriter();
    out.println("<html><body>");
    out.println("<h3>Preferences Saved Successfully!</h3>");
    out.println("<a href='DisplayServlet'>Go to Weather Page</a>");
    out.println("</body></html>");
}


}
```

3. DisplayServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class DisplayServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      Cookie cookies[] = request.getCookies();
```

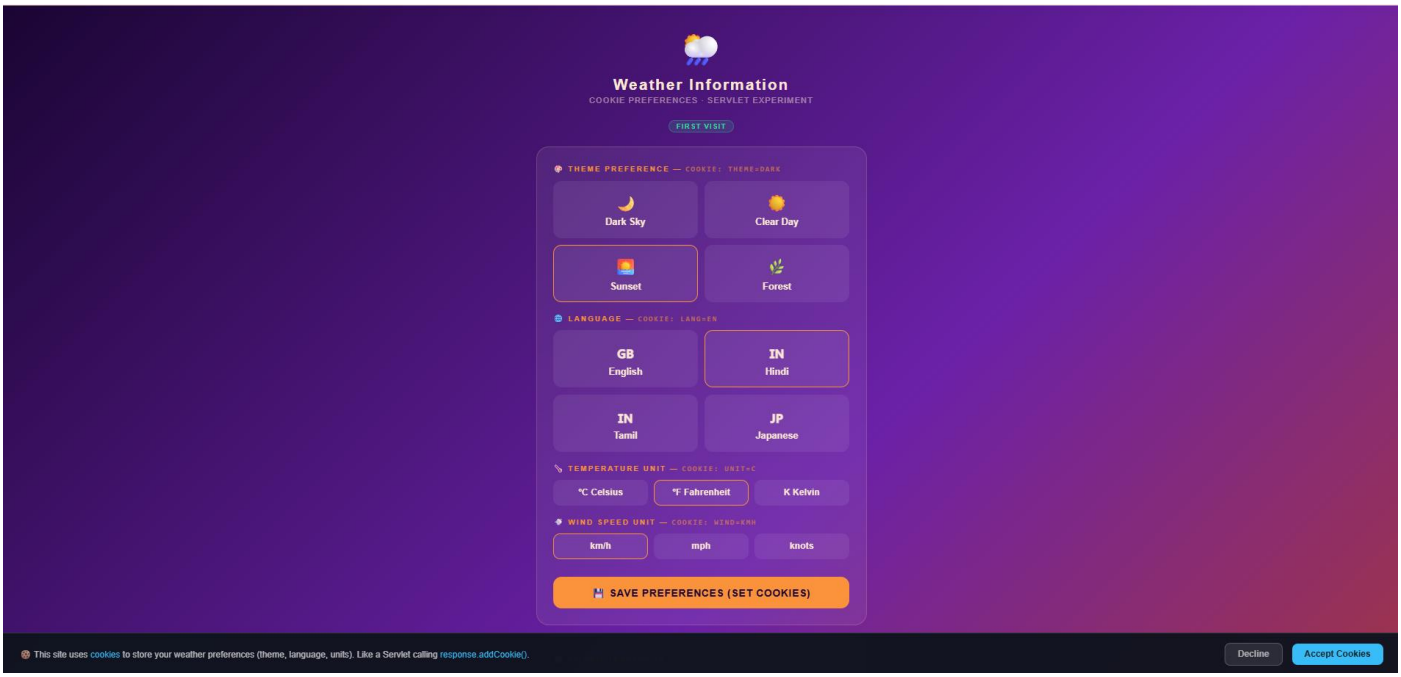
```
        String theme = "Not Set";
        String language = "Not Set";

        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) {
                    theme = c.getValue();
                }
                if (c.getName().equals("language")) {
                    language = c.getValue();
                }
            }
        }

        PrintWriter out = response.getWriter();
        out.println("<html><body>");

        // Apply theme
        if (theme.equals("Dark")) {
            out.println("<body style='background-color:black; color:white;'>");
        }

        out.println("<h2>Weather Information Page</h2>");
        out.println("<p><b>Theme:</b> " + theme + "</p>");
        out.println("<p><b>Language:</b> " + language + "</p>");

        // Dummy weather info
        out.println("<h3>Today's Weather</h3>");
        out.println("<p>Temperature: 32°C</p>");
        out.println("<p>Condition: Sunny</p>");

        out.println("<br><a href='index.html'>Change Preferences</a>");
        out.println("</body></html>");
    }
}
```

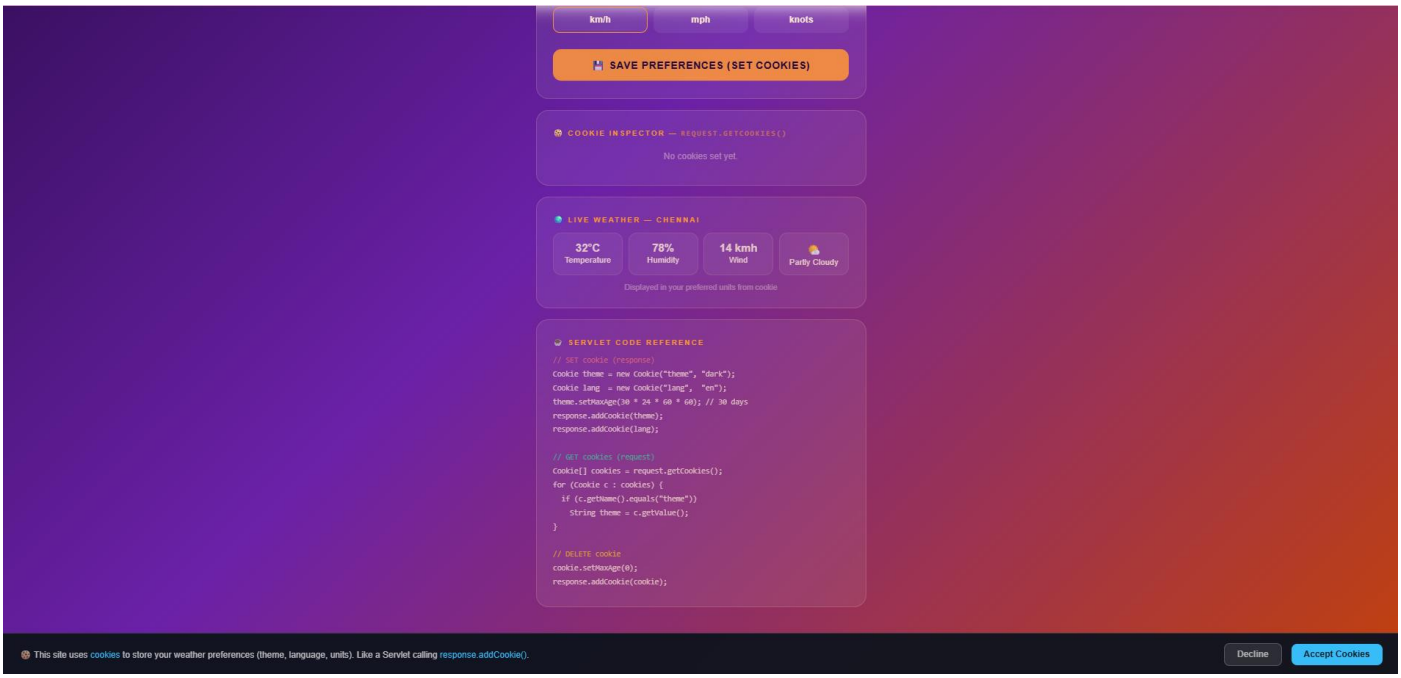
4. web.xml

```
<web-app>

    <servlet>
        <servlet-name>PreferenceServlet</servlet-name>
        <servlet-class>PreferenceServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>PreferenceServlet</servlet-name>
        <url-pattern>/PreferenceServlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>DisplayServlet</servlet-name>
        <servlet-class>DisplayServlet</servlet-class>
    </servlet>
```

```
    <servlet-mapping>
        <servlet-name>DisplayServlet</servlet-name>
        <url-pattern>/DisplayServlet</url-pattern>
    </servlet-mapping>

</web-app>
```

OUTPUT :





Result:

The experiment was successfully completed.User preferences (theme and language) were stored using cookies and retrieved on subsequent visits, allowing the weather webpage to provide a personalized user experience.